

Monitoring Log Files with Nagios Plugins



Plugins allow admins to monitor computer systems for errors in their functioning. The official Nagios package has 50 plugins to enable the monitoring of all basics. There are, however, more than 4000 Nagios plugins available to monitor every aspect of infrastructure. This article covers the installation of a plugin called *check_logfiles*.

The Nagios monitoring tool is one of the best open source solutions available to systems administrators to monitor servers and network elements. Using the Nagios framework, one can monitor servers, running processes, connected devices and network services. Also, it's easy to monitor custom applications.

This article focuses more on monitoring the log files using a Nagios plugin. We hope you have already set up Nagios core, Nagios plugins and NRPE (Nagios Remote Plugin Executer).

Some of the features of the Nagios monitoring tool include:

- Monitoring of network services (SMTP, POP3, HTTP, NNTP, PING, etc)
- Monitoring of host resources (processor load, disk usage, etc)
- A simple plugin design that allows users to easily develop their own service checks
- Support for implementing redundant monitoring hosts
- Contact notifications when service or host problems occur and get resolved (via email, SMS, or a user-defined method)

Overview of Nagios plugins

Unlike many other monitoring tools, Nagios does not include any internal mechanisms for checking the status of hosts and services on the network. Instead, it relies on external programs (called plugins) to collect the data.

Plugins are compiled executables or scripts that can be executed from the command line to check the status of a host or a service. Nagios uses the results from plugins to determine the current status of hosts and services on the network.

Nagios will execute a plugin whenever there is a need to check the status of a service or host. The plugin

performs the check and then simply returns the results for processing. Nagios will process the results that it receives from the plugin and take any necessary action (running event handlers, sending out notifications, etc).

Plugins as an abstraction layer

Plugins act as an abstraction layer between the monitoring logic present in the Nagios daemon and the actual services and hosts that are being monitored. Refer to Figure 1 for Plugins abstraction layer.

Available plugins

Currently, there are plugins available to monitor many different kinds of devices and services, including:

- HTTP, POP3, IMAP, FTP, SSH and DHCP
- CPU load, disk usage, memory usage, current users
- UNIX/Linux, Windows servers
- Routers and switches

Installing a plugin for log monitoring

Let's consider the SSH server installed at the client side for monitoring the log files using Nagios. There are several monitoring plugins available for Nagios, so let's go with the widely used *check_logfiles* to monitor the log files for the SSH server.

check_logfiles has some extra features compared to the default monitoring plugin:

- 1) It detects the log rotations and scans within the rotated archives as well.
- 2) More than one pattern can be defined, which can be classified as warning patterns and critical patterns.
- 3) *check_logfiles* lets you call scripts either after every hit, at the beginning or at the end of its runtime.